

their customers better by facilitating an observable context, knowledge about the world around and social values. Sensing is a crucial phenomenon associated with context awareness. Sensing can be external where visual tracking and location systems are used to collect context or internal wherein sensors are embedded within the user devices and generate context-rich information. Consider a scenario where a doctor has a device that senses its proximity to a particular patient and shows the patient records automatically, saving time for search and retrieval of such information. Or a device that automatically alerts you when it's about to rain and covers your clothes drying in the sun by a shade using humidity sensors. Or a device that detects your mood and changes the lighting of your room accordingly. Great minds are somewhere working in a lab to make all this possible for you.

Automatic contextual reconfiguration

Automatic Contextual Reconfiguration involves sensing the context and reacting in terms of either triggering a task or tweaking the settings of an existing system. Context-aware devices aim at imbibing a strong automatic contextual reconfiguration system. This phenomenon has a wide range of applications. Imagine how amazing it would be if your smartphone could sense when you're entering a meeting room/ICU and automatically put your phone on silent mode. Context reconfiguration is also very handy in security systems such as an application that disables cameras on all devices in a No Photography Zone. Another usage of context reconfiguration is in manufacturing industries to regulate the operation of machines. One such application is in the paper and pulp industry, where the lime kilns need continuous manual monitoring and adjustment of temperature for higher productivity. With embedded temperature sensors that automatically regulate temperature inside the kiln, it's possible to increase productivity and improve quality of the product by reducing manual intervention to check temperature.

Organic computing

Organic Computing adds an extra layer to context awareness and aims at developing intelligent systems that have self-X properties and systems that react to both, changes in itself (endogenous) and changes in the outside world (exogenous). Self-X properties of such systems include self-optimisation, self-protection, self-healing, self-support, self-configuration etc. Such devices behave independent of manual intervention as they can and use machine